

Surveillance Poisons Communication in LLM Global Games

WIP Seminar

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Motivation & Setup

The Information Channel Is a Trap

The same channel that helps citizens coordinate
is the channel regimes can **poison**.

Three hard problems in global games:

1. Lab experiments → stylized numeric settings, small n
2. Field data → confounded by institutions, unobserved information
3. Theory → tractable but hard to test at scale

This paper: Embed LLM agents in Morris–Shin (2003) regime-change game

- Private signals conveyed as **natural-language intelligence briefings**
- Pre-play communication on a network
- Surveillance, censorship, and information design treatments

This Talk

1. **Theory** — Global game of regime change (5 min)
2. **Design** — Signal $\rightarrow$ text $\rightarrow$ decision pipeline (10 min)
3. **Part I: Benchmark** — Do LLMs play threshold strategies? (10 min)
4. **Part II: Information Design** — Censorship & persuasion (10 min)
5. **Part III: Surveillance** — The central result (15 min)
6. **Discussion** (10 min)

Theory

The Global Game of Regime Change

Citizens $i \in [0, 1]$ choose: **JOIN** uprising ($a_i = 1$) or **STAY** home ($a_i = 0$)

- Regime strength $\theta \sim$ diffuse prior
- Regime falls $\iff A \equiv \int a_i di > \theta$

Payoffs:

$$u_i = \begin{cases} B & \text{join \& regime falls} \\ -C & \text{join \& regime survives} \\ 0 & \text{stay} \end{cases}$$

Private signals: $x_i = \theta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$

Equilibrium (Morris & Shin 2003)

Unique BNE in threshold strategies:

$$\text{Join} \iff x_i < x^* = \theta^* + \sigma \Phi^{-1}(\theta^*)$$

where $\theta^* = \frac{B}{B+C}$

Attack mass — theoretical join fraction at regime strength θ :

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right)$$

$\Rightarrow$ Decreasing sigmoid in θ : weaker regimes face larger uprisings

Empirical question: Do LLM agents approximate this?

Why LLMs?

What LLMs offer over lab experiments:

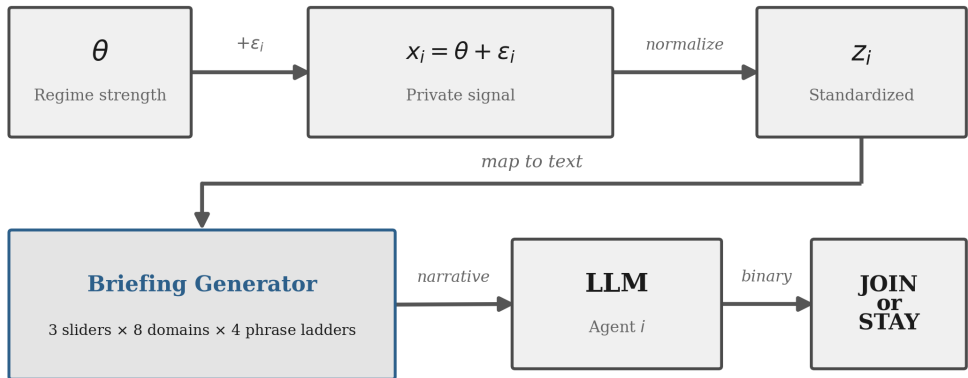
- Rich natural-language information environment (not just numbers)
- Scale: 7 models $\times$ 1,800 country-periods $\times$ 25 agents
- Full control over the information structure
- Identical prompt structure across models
- Replicable, cached, reproducible

What LLMs do *not* offer:

- Claims about human behavior
- Proof of “rational” reasoning
- The behavioral regularity is the *platform*, not the claim

Experimental Design

The Pipeline: Signal $\rightarrow$ Text $\rightarrow$ Decision



$\theta \rightarrow x_i = \theta + \varepsilon_i \rightarrow z_i = (x_i - \bar{z})/\sigma \rightarrow \text{briefing generator} \rightarrow \text{LLM call} \rightarrow \text{JOIN/STAY}$

The Briefing Generator

Core idea: Convert continuous z-scores into graded natural language

Three latent sliders control phrase selection across **8 evidence domains**:

Slider	Function	Effect
Direction	$\Lambda(0.8z)$ (logistic)	Pro-uprising vs. pro-regime tone
Clarity	$1 - e^{-(z/1.0)^2}$ (U-shaped)	Ambiguous near cutoff, clear at extremes
Coordination	$\Lambda(-0.6z)$ (decreasing)	Open coordination climate vs. suppressed

Domains: Elite cohesion, security posture, economic fundamentals, institutional resilience, public coordination, dissent signals, international stance, media environment

Example Briefings

$z = -2$ (weak regime)

PRIVATE BRIEFING

BOTTOM LINE: Signs of institutional erosion are accumulating faster than the regime can manage them.
ATMOSPHERE: Coordination is happening in plain sight – meetings, signals, preparations that would have been unthinkable a month ago.

EVIDENCE:

- [Elite Cohesion] senior figures are giving interviews that subtly distance themselves from the leader's recent decisions
- [Security Forces] reports of soldiers fraternizing with civilians more than usual at checkpoints (overwhelming convergence – every channel telling the same story independently)
- [Money And Logistics] customs and trade are flowing normally at the borders – no unusual delays or inspections (overwhelming convergence – every channel telling the same story independently)
- [Street Mood] jokes about the regime are being told openly in places where that would have been unthinkable
- [Information Control] leaked documents are circulating faster than the regime can discredit them (overwhelming convergence – every channel telling the same story independently)
- [Personal Observations] your cousin in the army hasn't called in two weeks – unusual for someone who calls every Sunday (near-unanimous confirmation from every reliable source)
- [Diplomatic Signals] international shipping companies have reclassified the country as higher-risk for cargo insurance – the trend cooling off, though not yet
- [Institutional Functioning] municipal services – garbage collection, water, permits – have become erratic in the capital

UNCERTAINTIES:

- Whether any of the apparent weakness is being deliberately staged to flush out opponents.
- How much of the disarray is structural versus a temporary response to a specific crisis.

$z = 0$ (ambiguous)

PRIVATE BRIEFING

BOTTOM LINE: The situation could tip either way – there are credible signals pointing in both directions.
ATMOSPHERE: There's a palpable sense that a critical mass is forming, even if nobody can count the numbers yet.

EVIDENCE:

- [Elite Cohesion] elites are publicly unified but privately positioning themselves for multiple outcomes (a couple of contacts mentioning it independently)
- [Security Forces] the security forces are maintaining presence but seem to be conserving resources rather than demonstrating strength (a couple of contacts mentioning it independently)
- [Money And Logistics] economic indicators are mixed – retail spending is down but construction permits are up
- [Street Mood] people are frustrated but fatalistic – complaining more, but not in a way that suggests action
- [Information Control] the regime is still shaping the narrative but has lost the initiative – they're responding to events rather than framing them
- [Personal Observations] a well-connected friend said 'it could go either way, and seemed genuinely unsure (a couple of contacts mentioning it independently)
- [Diplomatic Signals] foreign embassies are maintaining full operations but have updated contingency plans – standard prudence or genuine concern
- [Institutional Functioning] the system is creaking but not collapsing – people are working around the bottlenecks rather than through them (a couple of contacts mentioning it independently)

UNCERTAINTIES:

- Whether the population's quiescence reflects genuine acceptance or merely the absence of a triggering event.
- Whether the stability you're seeing is structural or just a temporary equilibrium that could shift quickly.

$z = +2$ (strong regime)

PRIVATE BRIEFING

BOTTOM LINE: Current indicators point to maintained control, with internal challenges being managed rather than spiraling.
ATMOSPHERE: People are aware of each other's frustrations but nobody is naming them directly. It's communication through omission.

EVIDENCE:

- [Elite Cohesion] the regime successfully managed a minor scandal by presenting a unified response within hours
- [Security Forces] security checkpoints have been reduced, suggesting confidence rather than siege mentality (overwhelming convergence – every channel telling the same story independently)
- [Money And Logistics] customs and trade are flowing normally at the borders – no unusual delays or inspections (overwhelming convergence – every channel telling the same story independently)
- [Street Mood] people complain but do it like people who expect tomorrow to look much like today
- [Information Control] social media monitoring appears sophisticated and well-resourced (overwhelming convergence – every channel telling the same story independently)
- [Personal Observations] your cousin in the army hasn't called in two weeks – unusual for someone who calls every Sunday (near-unanimous confirmation from every reliable source)
- [Diplomatic Signals] foreign students continue to enroll at local universities – their embassies haven't advised them to leave – the trend moving with a momentum that would be hard to reverse
- [Institutional Functioning] municipal services are running smoothly – a mundane but reliable indicator of institutional health

UNCERTAINTIES:

- Whether the regime's strength is being accurately perceived by potential challengers – overestimation could be self-fulfilling.
- If unrest flares, the response capacity seems intact – though no system is invulnerable to surprises.

Experimental Structure

Part	θ distribution	Baseline	Question
I (Benchmark)	$\mathcal{N}(\bar{z}, 1)$	Pure game	Threshold behavior?
II (Info Design)	Grid $[0.2, 0.8]$	Calibrated	Can we shift coordination?
III (Surveillance)	Same as I	Comm game	Can regimes exploit comms?

Seven models spanning six architecture families:

- Mistral Small Creative (primary), Llama 3.3 70B, Qwen3 30B/235B
- GPT-4o-mini-OSS, Trinity Large, MiniMax-M1

All: $n = 25$ agents, $\sigma = 0.3$, temperature = 0.7, via OpenRouter

One-degree-of-freedom calibration:

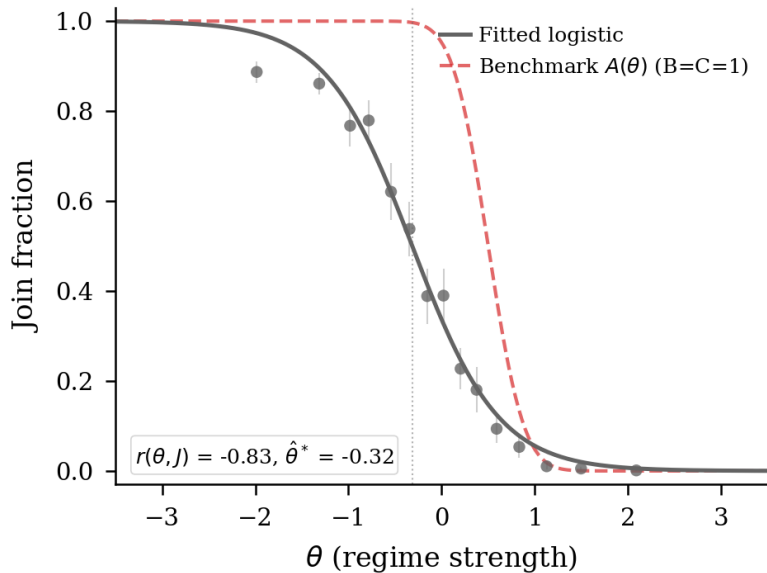
- Only adjust `cutoff_center` (shifts the z-score origin)
- Sigmoid **slope is emergent**, never optimized
- Iterative damped gradient until fitted logistic center ≈ 0

Key validation:

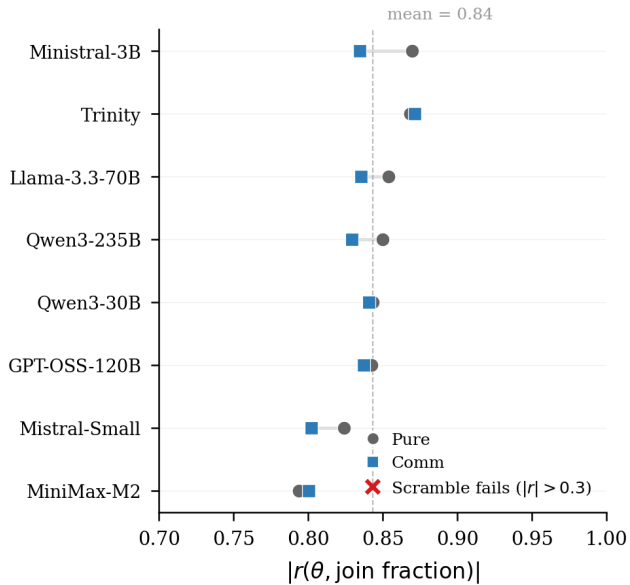
- Holdout validation confirms no overfitting
- 6 uncalibrated models still produce $|r| > 0.71$
- Deliberately miscalibrating (± 0.3 shift) leaves r unchanged
- $\Rightarrow$ Calibration adjusts the *level*, not the *slope*

Part I: Behavioral Benchmark

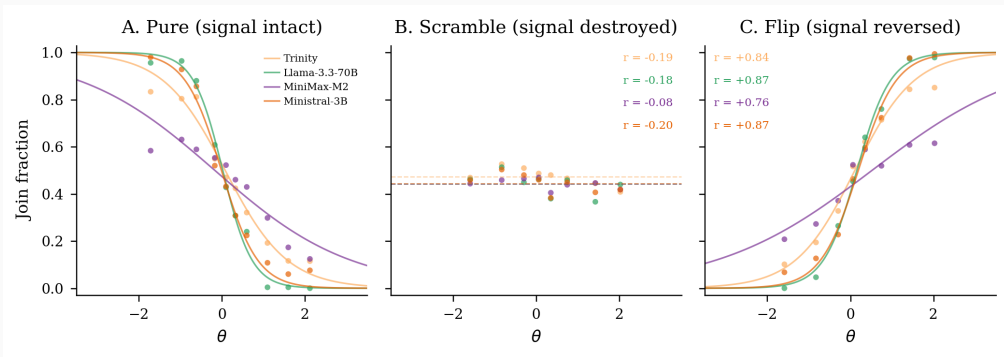
Result 1: Threshold-Policy Alignment



Cross-Model Replication



The Falsification Triptych



Pure: $r = +0.80$. **Scramble** (permute briefings across periods): $r = +0.03$. **Flip** (negate z-scores): $r = -0.80$. All replicate across 7 models.

Why Not Just Text Classification?

Three identification strategies beyond the sigmoid:

1. Narrative stakes ladder (fixed briefing body, vary header only):

- 8 conditions from existential consequences ' ' to zero cost ' '
- Mean joining: 21% $\rightarrow$ 79% (57.6 pp range)
- Placebo (weather) = 33% — irrelevant context dilutes signal
- Text classifier sees only briefing body $\Rightarrow$ cannot explain the shift

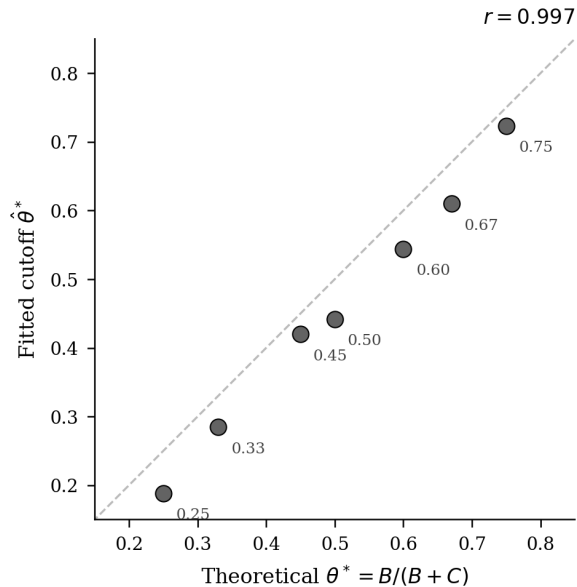
2. Payoff sweep (B/C ratio varies θ^*):

- 7 conditions, fitted cutoffs track θ^* with $r = 0.997$
- Pipeline centering diagnostic, not payoff-reasoning proof

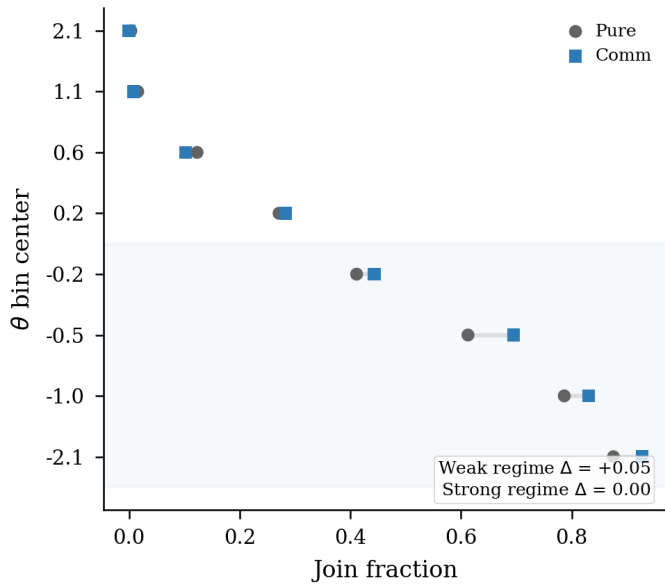
3. BNE vs. Level- k :

- BNE fits better (RMSE = 0.199) than L_1 (0.307) or L_2 (0.348)
- But: centering diagnostic, not clean L_k rejection

Narrative Stakes Ladder



Communication Creates an Informative Channel



Part II: Information Design

Bayesian persuasion (Kamenica & Gentzkow 2011; Goldstein & Huang 2016):

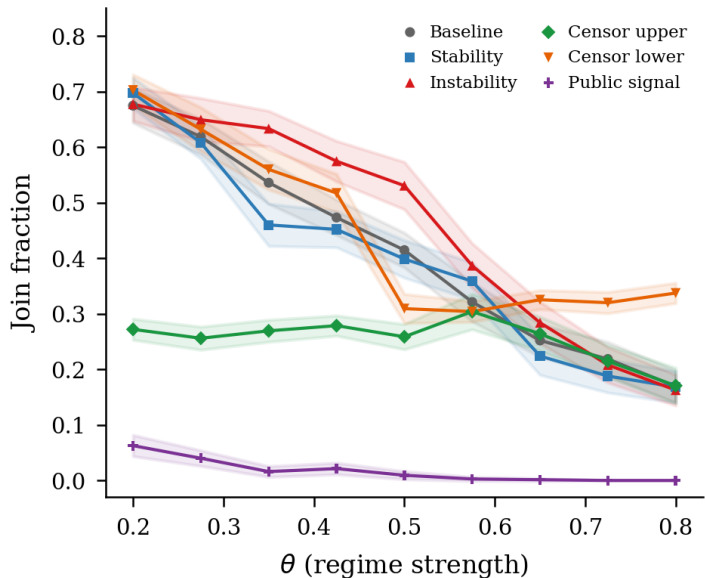
The designer controls $\pi : \Theta \rightarrow \Delta(\mathcal{S})$ — the mapping from states to signal distributions

Implementation: Modify the briefing generator's slider functions near θ^*

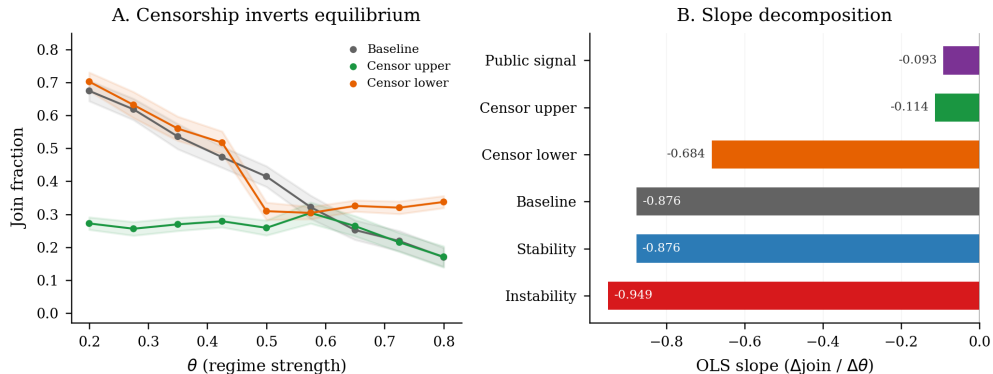
$$w(\theta) = \exp\left(-\left(\frac{\theta - \theta^*}{h}\right)^2\right)$$

Design	Manipulation	Prediction
Stability	Widen clarity kernel	Suppress joining
Instability	Sharpen signals	Raise joining
Upper censorship	Pool $\theta \leq \theta^*$ to neutral	Suppress joining
Lower censorship	Pool $\theta \geq \theta^*$ to neutral	Attenuate slope
Public signal	Add shared observations	Crowd out private info

All Information Designs



Censorship: Upper vs. Lower



Upper censorship (pool weak-regime states): creates flat “plateau” — agents can’t distinguish $\theta = 0.20$ from $\theta = 0.50$. Join rate drops to 25.4%.

Lower censorship (pool strong-regime states): attenuates slope, raises mean joining to 44.5%.

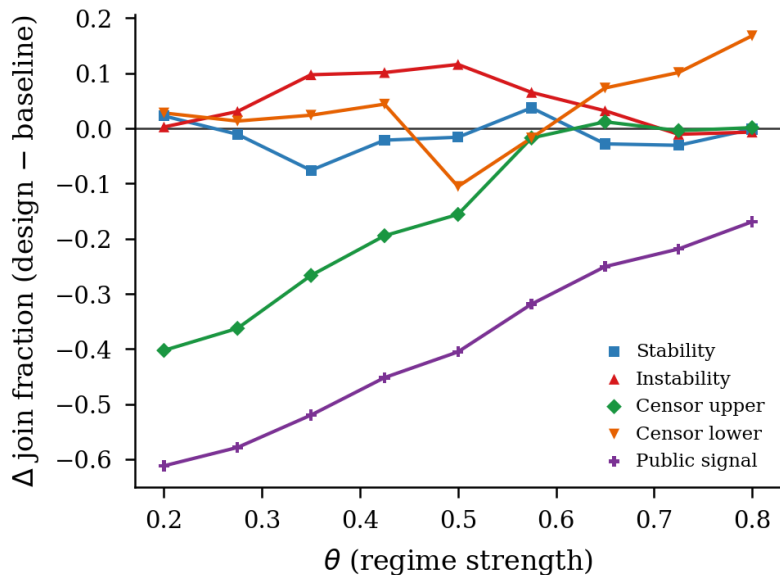
Censorship with Common Knowledge

When agents are told the regime censors unfavorable intelligence:

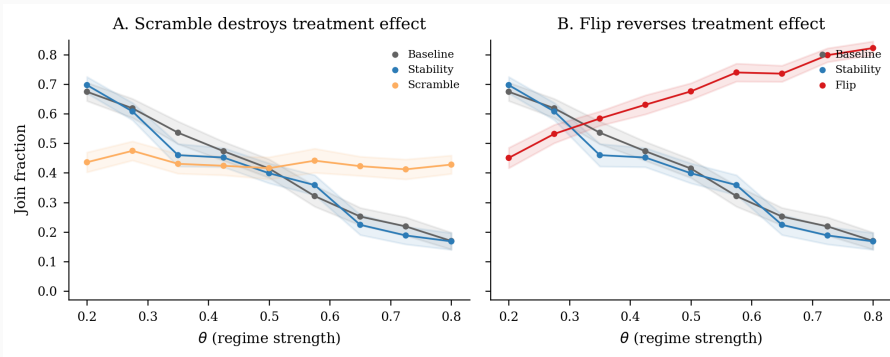
Condition	Mean join	Δ vs. baseline
Baseline	40.9%	—
Naïve upper censorship	25.4%	−15.4 pp
Common-knowledge censorship	43.2%	+2.4 pp

- Common knowledge **largely neutralizes** mean suppression
- But signal-responsiveness stays attenuated: $r = +0.74$ vs. $+0.88$ baseline
- Can't undo information *loss* even when you know about it

Treatment Effects by θ

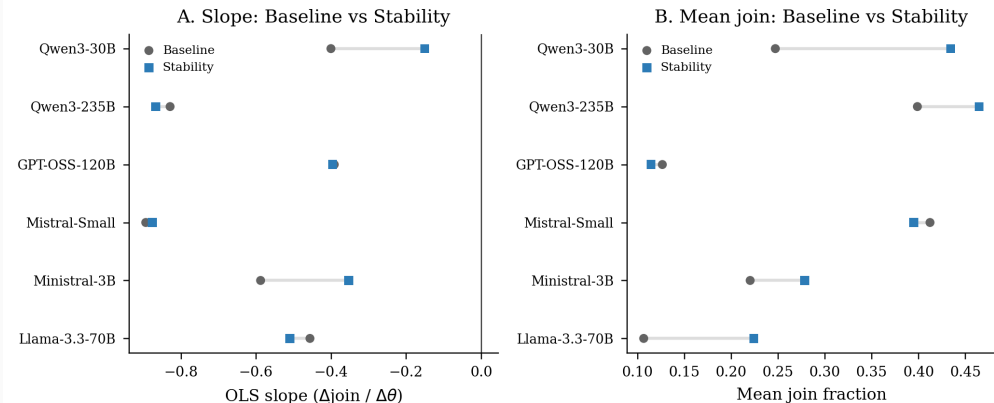


Identification Within Information Design



Scramble: $r = +0.11$ (vs. baseline $+0.89$). Flip: $r = -0.82$. Hard scramble (all briefings from $\theta = 0.5$): $r \approx 0$ for both Mistral and Llama.

Cross-Model Replication



Flip inversion replicates across all five models. Scramble collapse is clean for 3/5 models; Llama retains residual ecological signal from within-country permutation (hard scramble confirms $r \approx 0$).

Part III: Surveillance (The Central Result)

Architecture is key:

1. **Communication stage:** Surveillance warning appended to message prompt

“You have reason to believe that your communications are being monitored by regime security services. Messages deemed subversive could have serious consequences.”

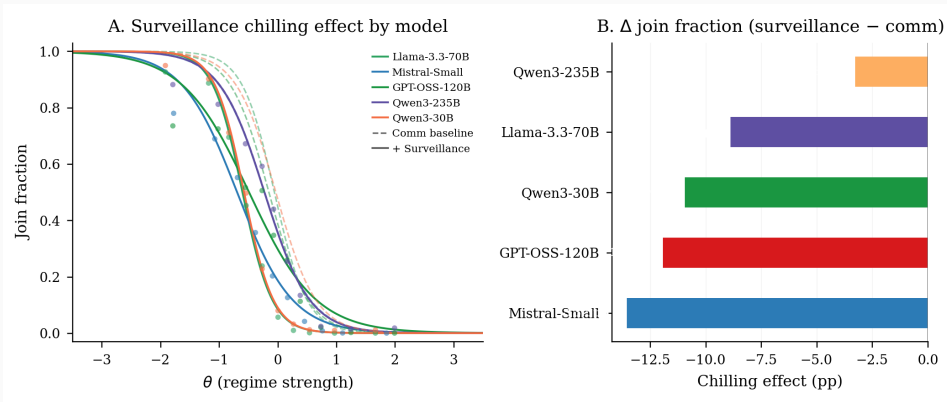
2. **Decision stage:** Standard prompt, **no surveillance reference**

Each LLM call is stateless — no memory across calls

⇒ Any effect must operate through the *content of messages* the decision stage receives

⇒ **Indirect channel:** surveillance → self-censored messages → degraded information → lower joining

Result: Surveillance Produces a Large Chilling Effect



—9.7 pp average across five models. Mistral: —13.5 pp. Preserved r : surveillance is a level shift, not disruption of signal processing.

The Belief–Action Wedge

Actions shift; beliefs don't:

	Beliefs	Actions
Pure	44.4%	41.4%
Comm	44.4%	41.4%
Surveillance	43.6%	28.5%
Δ (surv vs. pure)	−0.7 pp $p = 0.18$	−12.9 pp $p < 0.001$

Second-order beliefs also barely move:
31.2% $\rightarrow$ 30.9% ($p = 0.59$)

Interpretation caveat:

Beliefs are post-decision scalar readouts on a restricted info set (no peer messages).

$\Rightarrow$ “Wedge between actions and *elicited beliefs*”, not proof that full strategic state is unchanged.

Consistent with Kuran's (1991) preference-falsification logic, while remaining agnostic about mechanism.

How Self-Censorship Works

Message informativeness drops 81%:

$$R^2(\theta \mid \text{messages}) : 0.12 \rightarrow 0.02$$

Word-frequency evidence:

Word/phrase	Communication	Surveillance
act'' 15.8\% 4.1\% collapse''	9.6%	0.6%
Action-signaling (among JOINers)	80.0%	61.0%

⇒ Agents who privately decide to JOIN still self-censor their messages

The Degraded-Messages Control

Key identification test: Replace all LLM messages with generic text

(I'm still assessing the situation, ''It's hard to say what will happen'')

- No surveillance framing, same two-stage protocol
- Join rate: 41.1% $\rightarrow$ 20.2% (-20.9 pp, $t = -6.89$, $p < 0.001$)

Interpretation:

- Degraded messages produce *larger* suppression than surveillance (-20.9 vs. -13.5 pp)
- Expected: degraded removes *all* information; surveillance only partially degrades
- $\Rightarrow$ Main mechanism is **communication poisoning**, not direct deterrence

Placebo Tests

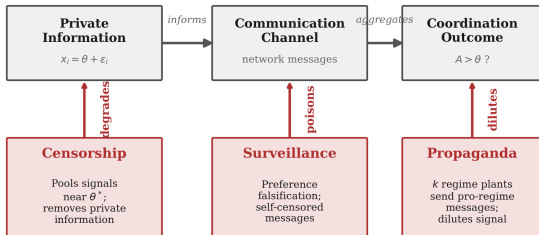
Variant	Δ vs. baseline	<i>p</i> -value
Monitored for research'' (no consequences) \$+2.6\$ pp \$0.416\$ Aggregated anonymously''	+4.0 pp	0.201
Regime surveillance	−13.5 pp	< 0.001

⇒ The chilling effect is **specific to the surveillance warning's strategic content**

Not just someone is watching'' --- it's someone with power to punish is watching''

Surveillance × Censorship: Complements

The Information Channel as a Vulnerability



Instrument Interactions

Surv. + Propaganda:

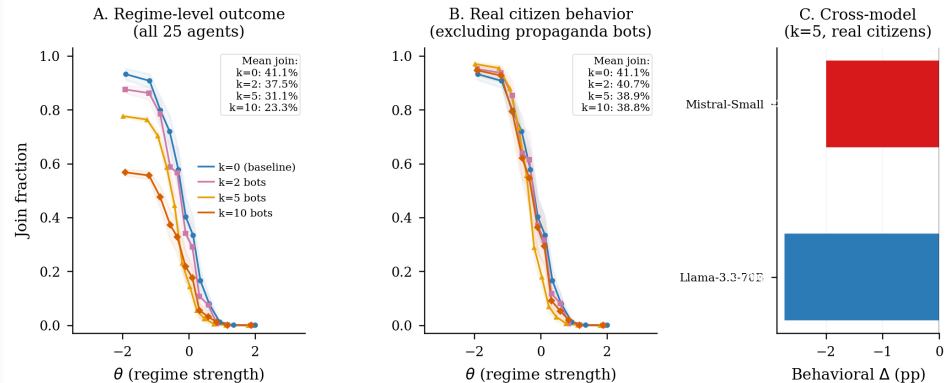
Approximately additive

Surv. + Censorship:

Super-additive: -11.4 pp under censorship vs -2.1 pp at baseline

The regime does not need to change what citizens believe; it needs only to make them uncertain about each other.

Propaganda: A Weaker Instrument



m regime plants (always STAY) in the 25-agent network. Overall join rate drops mechanically, but **behavioral effect on real citizens saturates quickly**: -0.4 pp ($m = 2$), -2.0 pp ($m = 5$), -2.3 pp ($m = 10$). Doubling plants from 5 to 10: $\Delta = -0.3$ pp ($p = 0.67$).

Robustness & Extensions

Three tests that the sigmoid isn't mechanical:

1. Alternative direction transforms:

- Tanh: $r = 0.83$, Linear: $r = 0.83$, Step: $r = 0.84$ (baseline logistic: $r = 0.75$)
- Aggregate sigmoid not tied to any particular mapping

2. Cross-generator robustness:

- Three prose formats (narrative, diplomatic cable, journalistic wire)
- Max within-model r difference: 0.02

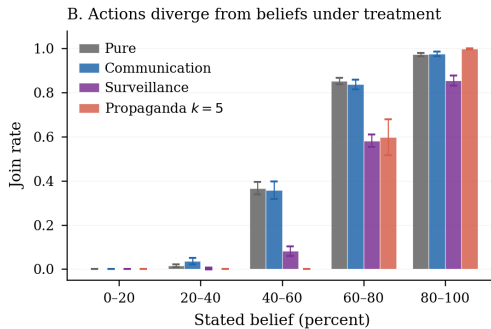
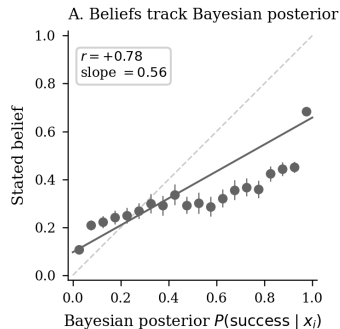
3. Domain ablation:

- Remove coordination domains $\Rightarrow r = 0.90$ (preserved)
- Remove state domains $\Rightarrow r = 0.86$ (attenuated)
- Signal distributed across all 8 domains

Additional Robustness

Test	Result
Temperature ($T \in \{0.3, 0.5, 0.7, 1.0, 1.2\}$)	$r \in [+0.76, +0.84]$ across 13 combos
Agent count ($n \in \{5, 10, 50, 100\}$)	$r \in [+0.60, +0.65]$
Network density ($k = 4$ vs. $k = 8$)	$r = +0.74$ vs. $+0.66$
Mixed-model populations	$r = +0.77$ (pure), $+0.75$ (comm)
Uncalibrated models (6 tested)	All $ r > 0.71$, 5 exceed 0.75
Placebo calibration (± 0.3 shift)	r unchanged
Text baseline ($\hat{p} = 1 - d$)	$r = 0.79$ but LLM sigmoid is steeper

Beliefs Track Theory



Stated beliefs vs. theoretical posterior ($r = +0.78$). Beliefs predict actions ($r = +0.84$). But: post-decision elicitation $\Rightarrow$ near-perfect pseudo- R^2 (0.975) likely reflects rationalization.

Discussion

Central result: Surveillance poisons communication

- -9.7 pp across 5 models
- Message informativeness drops 81%
- Beliefs barely move; actions shift sharply
- Degraded-messages control: -20.9 pp (larger than surveillance)
- Placebo: no effect
- $\Rightarrow$ Mechanism is communication poisoning, not deterrence

Supporting results:

- Robust threshold behavior across 7 architectures ($\bar{r} = +0.80$)
- Information design shifts coordination in theory-consistent directions
- 57.6 pp range from narrative stakes alone
- Censorship + surveillance are complements; propaganda saturates

1. **Emergent strategic self-censorship:** - No training signal for deception needed - Capability emerges from pretraining on human strategic text
2. **Belief monitoring limitations:** - Post-decision beliefs strongly predict behavior (pseudo $R^2 = 0.975$) - But alignment techniques monitoring only inputs/outputs may miss decision-relevant computation
3. **Information-structural manipulation:** - Up to 40 pp behavioral shift from information design - Context fields that body-only monitors ignore - Adversarial prompt design can exploit this

Limitations

1. **Mechanism ambiguity:** Approximate Bayesian reasoning or learned heuristic that matches BNE? Behavioral claim doesn't require resolving this.
2. **Static design:** Real coordination problems are dynamic (Angeletos et al. 2007, Chassang 2010). Persistent surveillance may produce adaptation.
3. **External validity:** "Homo silicus" (Horton 2023). No claim that LLMs predict human behavior. What they offer: a *calibrated testbed* for information-structure comparative statics.
4. **Belief elicitation:** Post-decision, restricted information set, scalar readouts. Interpret as descriptive, not as validated mediator.

- **Belief elicitation on full information set** (include peer messages)
- **Within-batch robustness** for auxiliary designs
- **Dynamic extensions** — communication poisoning over time
- **Richer communication** — multi-round, more realistic networks
- **Human replication** — do the comparative statics transfer?

Paper, code, data, cached LLM responses:
<https://github.com/keltokhy/llm-global-games>

Thank you

Questions?

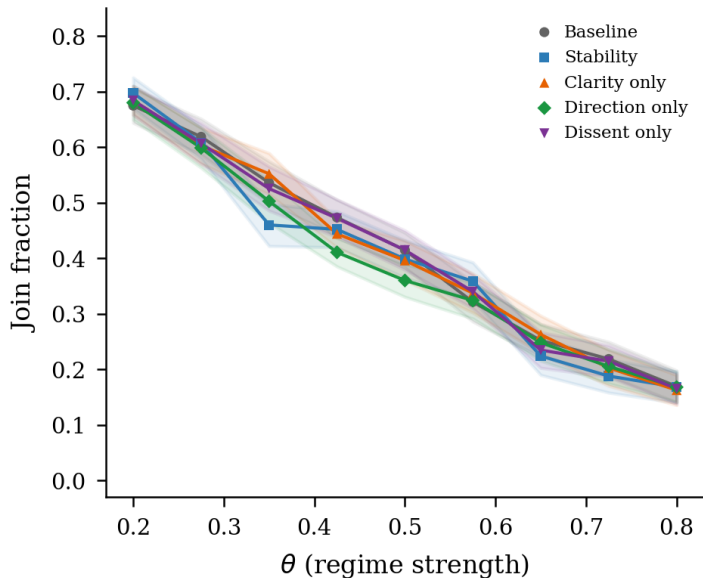
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<https://github.com/keltokhy/llm-global-games>

Backup Slides

Decomposition: Which Channel Drives Stability?



Publicness × Authority Factorial

	High coordination	Low coordination
Public (CK)	42.7%	43.5%
Private	37.8%	37.4%

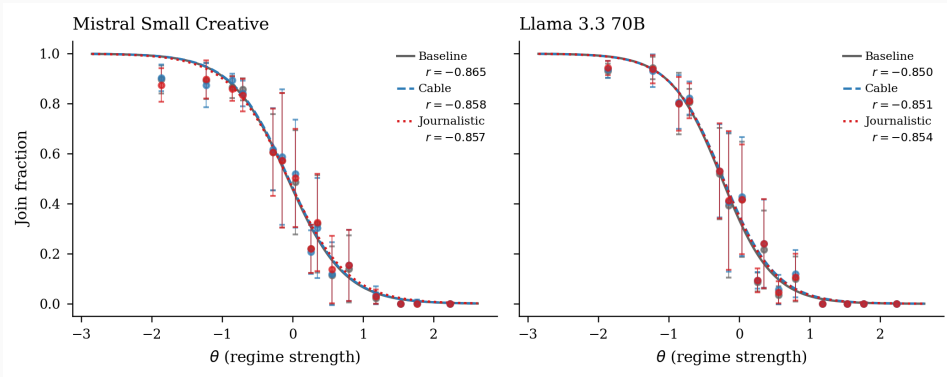
- Public framing main effect: +6.1 pp ($p = 0.0003$)
- Interaction (public × coordination): -1.2 pp ($p = 0.62$)
- Authority matters only under public framing (+5.4 pp public vs. -1.7 pp private)
- ⇒ Consistent with strategic complementarity, not just authority cuing

Finite- n Aggregation Check

With $n = 25$ discrete agents: regime falls when $\sum a_i > n\theta$

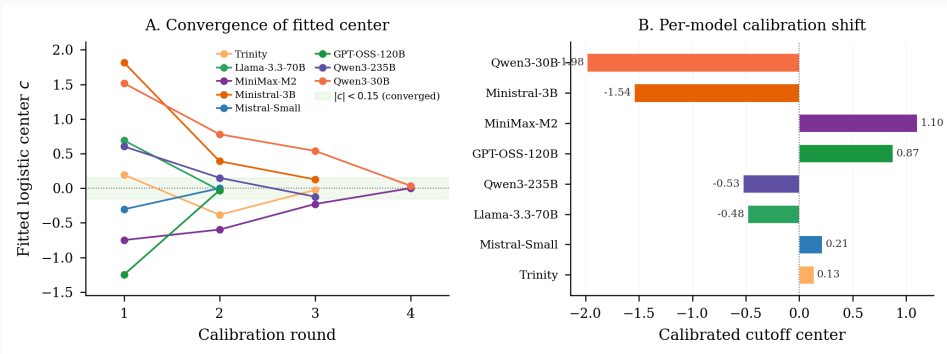
- Fit logistic on 70% train, predict $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$
- Predicted regime-fall rates correlate with empirical at $r \geq 0.95$ for every model
- Mistral: $r = 0.997$
- $\Rightarrow$ Binomial aggregation from fitted individual probabilities tracks realized outcomes

Cross-Generator: Three Prose Formats



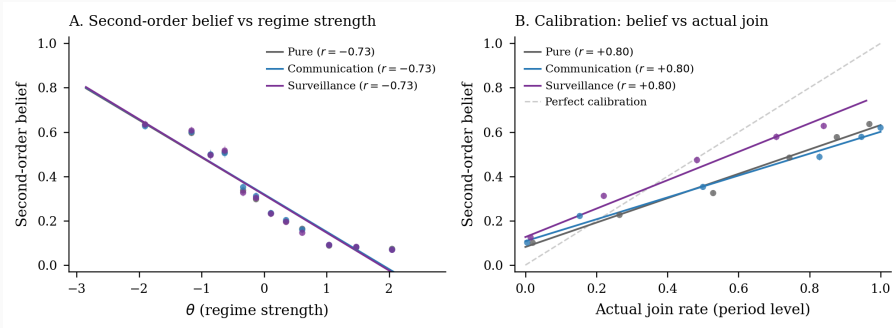
Baseline narrative, diplomatic cable, journalistic wire. Identical slider functions, different surface prose. Max within-model r difference: 0.02. Prose style is not the driver.

Calibration Convergence



Left: Trajectory of fitted logistic center across autocalibration rounds. *Right:* Final calibrated cutoff center per model. Cross-model heterogeneity in calibration quality (r_A from mid-0.6s to mid-0.8s).

Second-Order Beliefs Under Surveillance



Second-order beliefs (predictions about others' join rates): 31.2% $\rightarrow$ 30.9% under surveillance ($p = 0.59$). Agents change what they *do* without changing what they *state* about others.

$N = 289,555$ agent-level observations, clustered SEs

Marginal effects at the mean:

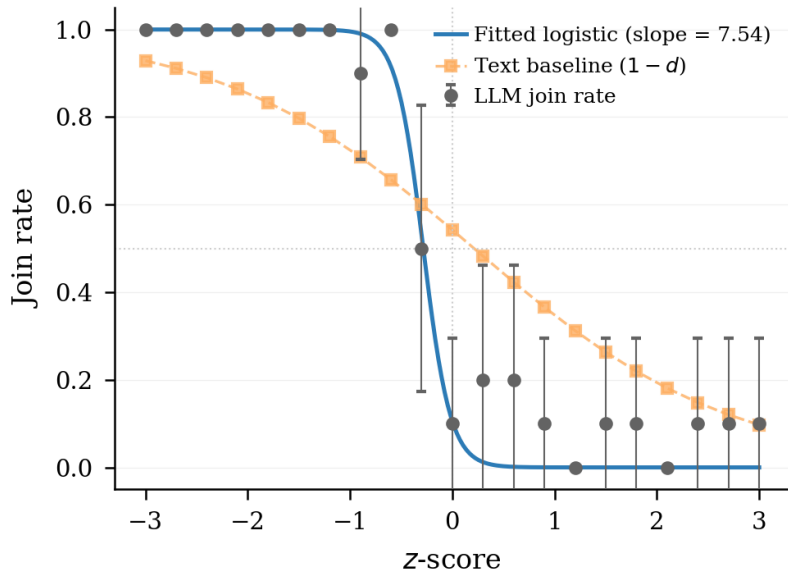
- 1-unit increase in θ : -41 pp on $P(\text{join})$
- Surveillance treatment: -36 pp on $P(\text{join})$

Main logit: Pseudo $R^2 = 0.381$

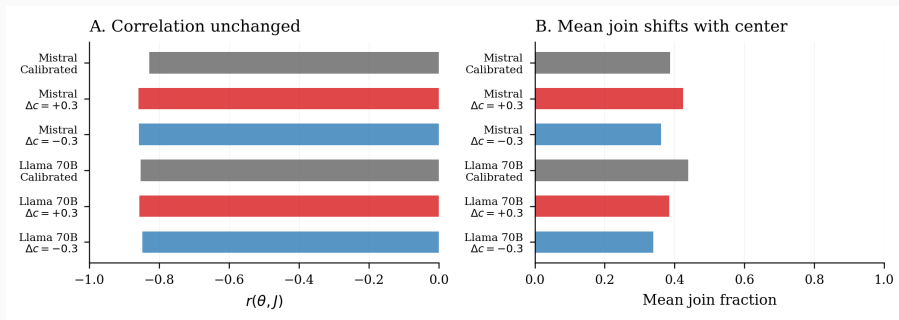
With beliefs: Pseudo $R^2 = 0.975$

Caveat: beliefs elicited post-decision $\Rightarrow$ near-perfect fit likely reflects rationalization, not causal mediation.

Text Baseline Comparison



Placebo Calibration



Deliberately miscalibrating Mistral and Llama (± 0.3 shift) leaves r virtually unchanged. Calibration adjusts the *level* but does not create the *slope*.